MVCH: Multiview Collaborative Hybrid Network for Hyperspectral Image Classification

Shujie Ding, Xiaoli Ruan, Jing Yang, *Member, IEEE*, and Zhiqiang Du

***Abstract*—Although deep learning methods have achieved significant progress in hyperspectral image (HSI) classification, existing approaches typically focus on modeling a single dependency view, such as local spatial–spectral, structural relational, or global contextual view, making it difficult to coordinate complementary views within a unified representation space. To address this issue, a multiview collaborative hybrid network (MVCH) is proposed for HSI classification. MVCH integrates multiple dependency views within a unified representation framework. Specifically, the graph-depthwise cooperative (GDC) module collaboratively models structural relational and local spatial–spectral views to enhance feature discriminability for fine-grained variations and discontinuous structural regions. Furthermore, the context-interaction Transformer (CIT) collaboratively models global contextual, local perceptual, and dynamic dependency views, enabling effective contextual interaction and adaptive dependency learning in complex HSI scenes. Extensive experiments conducted on three representative urban and agricultural HSI datasets demonstrate that MVCH effectively improves feature representation capability in complex scenes through unified multiview collaborative feature learning, achieving competitive classification performance compared with existing representative methods. The code is available at https://github.com/dsj0507/MVCH**

***Index Terms*—Deep learning, feature extraction, hyperspectral image (HSI) classification, Transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) is a high-resolution remote sensing modality in which each pixel contains continuous reflectance information spanning dozens to hundreds of narrow spectral bands. This distinctive property enables HSIs to finely discriminate ground objects based on their unique spectral signatures, making it an indispensable technology in applications such as forestry monitoring, geological exploration, and marine conservation.

Deep learning methods have achieved significant improvements in HSI classification by learning discriminative feature representations. Convolutional neural networks (CNNs) have become one of the most widely adopted frameworks owing to their strong local feature extraction capability and translation invariance. By exploiting local spatial neighborhoods, CNN-based methods can effectively capture local spatial–spectral details. However, their inherently limited receptive fields restrict the modeling of long-range contextual dependencies, reducing their capability to represent complex ground object categories. As illustrated in Fig. 1(a), the learned features exhibit noticeable interclass overlap and blurring, indicating limited discriminative ability. To overcome the local limitations of CNNs, Paoletti et al. [1] developed a pyramid residual network that improves network performance by gradually increasing the diversity of spatial–spectral attributes across levels. In addition, some researchers [2], [3] have introduced attention mechanisms to highlight important spatial regions and specific spectral bands, enhancing useful information and suppressing useless information. Although these methods have brought improvements, they have not changed the local nature of convolution operations.

To overcome the locality limitation of convolution operations, the Vision Transformer (ViT) introduces a self-attention (SA) mechanism to establish global interactions among image tokens and has achieved impressive performance in various visual tasks [4], [5]. Several studies [6], [7] have further developed hybrid CNN–Transformer architectures to simultaneously exploit local and global information. As shown in Fig. 1(b), Transformer-based representations exhibit improved intraclass compactness and interclass separability compared with CNN-based methods. Nevertheless, most existing hybrid architectures rely on simple module stacking and lack a systematic mechanism for coordinating local feature perception and global contextual modeling, which may lead to insufficient interaction between different feature streams.

Graph convolutional networks (GCNs) provide another perspective for HSI representation learning by explicitly modeling structural relationships among pixels. Through graph construction and neighborhood aggregation, GCN-based methods can effectively capture nonlocal contextual dependencies and irregular spatial structures. Consequently, they have attracted considerable attention in HSI analysis. However, existing graph-based approaches [8], [9] primarily focus on relational information propagation while lacking an explicit mechanism to characterize regular local spatial–spectral variations. As a result, the learned representations may suffer from limited discriminability. This phenomenon can be observed in Fig. 1(c), where graph-based representations exhibit relatively weak intraclass compactness and insufficient interclass separability.

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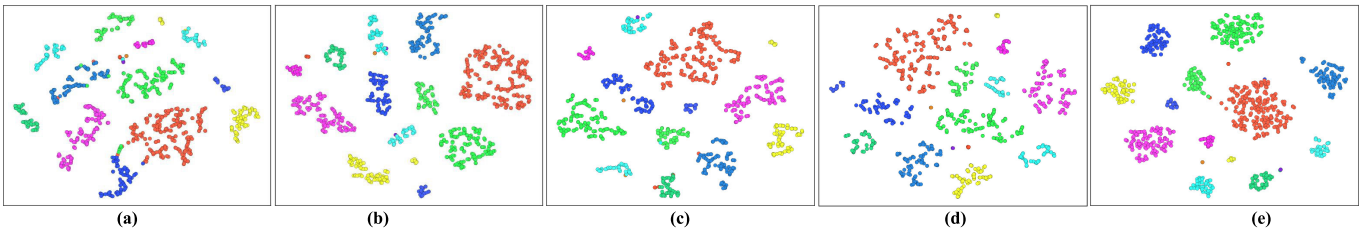


Fig. 1. T-SNE visualization of learned feature representations produced by different hyperspectral feature modeling paradigms. (a) CNN-based. (b) Transformer-based. (c) GCN-based. (d) SSM-based. (e) Ours.

Recently, state-space model (SSM)-based architectures represented by Mamba have demonstrated promising performance in HSI classification by efficiently modeling long-range dependencies with linear computational complexity. Benefiting from their sequence modeling capability, several SSM-based methods have achieved encouraging results in spatial–spectral representation learning. However, most existing SSM-based approaches primarily focus on sequential dependency modeling and usually operate within a single representation paradigm, while the complementary characteristics of local spatial–spectral details and structured contextual relationships remain underexplored. Consequently, the learned representations still exhibit noticeable intraclass dispersion and interclass ambiguity, as shown in Fig. 1(d).

The above observations reveal a common limitation of existing methods. Although CNN-, Transformer-, GCN-, and SSM-based approaches excel at modeling different types of dependencies, they typically emphasize a specific representation perspective and fail to effectively coordinate local details, structured relationships, and global contextual information within a unified representation space. Consequently, the complementary advantages of different feature modeling paradigms cannot be fully exploited. To address this issue, a multiview collaborative hybrid network (MVCH) is proposed for HSI classification. Different from existing methods that focus on a single dependency perspective, MVCH establishes a unified multiview collaborative representation framework by jointly modeling local spatial–spectral details, structured contextual relationships, and global dynamic contextual dependencies. As illustrated in Fig. 1(e), the proposed MVCH achieves significantly improved intraclass compactness and interclass separability, demonstrating the effectiveness of multiview collaborative representation learning.

Specifically, graph-depthwise cooperative (GDC) is designed to jointly model structural relational and local spatial–spectral views, where structural relationships provide relational constraints for local feature representation while local spatial–spectral information preserves fine-grained feature variations. Meanwhile, the context-interaction Transformer (CIT) integrates global contextual, local perception, and dynamic dependency views, where the global–local combined attention (GLCA) mechanism enhances contextual interaction with spatial-aware perception, while dual-path state fusion (DPSF) captures dynamic dependencies with complementary spatial feature enhancement. Through the interaction of these complementary views, MVCH enables different dependency information to be jointly modeled within a unified representation space. The main contributions of this work are summarized as follows.

1) MVCH is proposed to establish a unified multiview representation framework for the collaborative representation of complementary dependency information.
2) GDC is proposed to combine graph relational learning with depthwise feature refinement, facilitating interaction between structural relational and local spatial–spectral information and enhancing discriminability for fine-grained variations and discontinuous structural regions.
3) CIT is proposed to enhance contextual interaction and dynamic dependency representation. Specifically, GLCA introduces local perception into global interaction while preserving local spatial information, whereas DPSF employs a dual-path state-space mechanism to capture dynamic long-range dependencies.
4) A series of experiments on three HSI datasets covering both urban and agricultural scenes demonstrates that MVCH achieves more stable and competitive classification performance compared to representative CNN-, Transformer-, GCN-, and SSM-based methods.

II. Related Work

A. Transformer-Based HSI Classification

Recently, the Transformer architecture has attracted increasing attention, as its global SA alleviates the limitations of CNNs in modeling long-range dependencies. SA facilitates long-range token interactions, expanding the receptive field and strengthening global feature modeling. Consequently, ViT-based approaches have been increasingly explored for HSI classification [10], [11]. For example, Liao et al. [12] constructed separate spatial and spectral Transformer modules to capture long-range dependencies. In [13], a dual-branch Transformer is introduced to improve accuracy by jointly extracting and fusing spatial–spectral representations. Although these methods advance global feature modeling, the sequence-based processing of ViT still limits the capture of local spatial structures and fine-grained details. To mitigate this limitation, several studies incorporate local feature modeling into the ViT framework to balance global context modeling with local detail preservation. Notably, Huang et al. [14] proposed a 3-D Swin Transformer with a hierarchical windowing mechanism tailored to the 3-D spatial–spectral characteristics of HSIs, improving multiscale feature representation. Song et al. [15] proposed the interactive spatial–spectral fusion Transformer

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(ISSFormer), which integrates SA with convolution in a parallel design to simultaneously capture local spatial features and global spectral features. Niu et al. [16] combined a Transformer and CNN to extract global features in both spectral and spatial dimensions.

Although these approaches enhance local modeling to varying degrees, most rely on additional branches, windowing strategies, or simple parallel designs, resulting in complex architectures and weak integration between global attention and local modeling. To address this issue, a GLCA module is proposed by revisiting the internal structure of SA. By introducing lightweight local modeling operations into attention computation, GLCA tightly integrates global context modeling with local spatial structure learning while retaining the benefits of global modeling.

B. GCN-Based HSI Classification

GCN effectively captures nonlocal similarities among samples by constructing graph structures and aggregating information between nodes, demonstrating significant advantages in modeling complex spatial relationships. As a result, GCN methods have attracted extensive attention in hyperspectral data analysis [17]. Recently, researchers have proposed a variety of improved GCN-based approaches for HSI classification, with emphasis on graph construction, relational modeling, and feature fusion. Chen et al. [18] proposed the spatial–spectral dynamic GCN, which dynamically updates node similarities by integrating graph embeddings from the previous layer with current spatial–spectral node features. Ding et al. [19] adopted an adaptive smoothing operation based on the Kullback–Leibler divergence theory to maintain the diversity of adjacent nodes and enhance the quality of the graph structure. To enhance correlation modeling across spectral bands, Zheng et al. [20] proposed a graph-Transformer module that effectively integrates local neighborhood features with global long-range dependency features. In addition, Lu et al. [21] combined CNN with prototype learning to design a separable deep GCN, which extracts regional-level global structural information under limited parameter scales.

However, HSI typically contains hundreds of contiguous spectral bands and exhibits strong sequential characteristics and local continuity in spatial–spectral features [22]. Although existing GCN-based methods have achieved promising performance through adaptive graph construction, relational propagation, and the integration of auxiliary spatial–spectral features, they mainly focus on improving graph representation or enhancing specific feature fusion strategies. Considering that structural relational information and local spatial–spectral variations characterize complementary dependency patterns, effectively coordinating these views within a unified representation space provides an important direction for further improving HSI representation. To address this issue, a GDC module is proposed to establish a structural relational view through graph-based dependency modeling and coordinate it with a local spatial–spectral view within a unified interaction framework. By collaboratively exploiting these complementary views, GDC captures both structural dependencies and local feature variations, enhancing discriminative representation for complex HSI scenes.

C. State-Space-Model Based HSI Classification

Recently, SSMs [23], [24] have attracted considerable attention due to linear computational complexity and efficient modeling capability for long-range dependency capture. The Mamba model proposed by Gu and Dao [25] has emerged as a representative work in SSM research, characterized by concise structural design and strong performance in long sequence modeling. Subsequently, researchers further explored the application of Mamba in visual tasks such as image classification and proposed various improvements to enhance modeling capability and task adaptability [26], [27]. Several studies have also begun incorporating Mamba into HSI classification tasks [28], [29]. As an example, Ahmad et al. [30] introduced the morphological space Mamba and spectral morphological space Mamba models named morpmamba, providing computationally efficient alternatives to Transformer-based approaches. Zhang et al. [31] proposed the CenterMamba network, which achieves spatial–spectral long-range dependency modeling with linear complexity and improves computational efficiency. Zhou et al. [32] introduced the Mamba in Mamba framework, demonstrating the potential of Mamba for hierarchical feature modeling through multilevel representation learning. To further strengthen spatial modeling capability, Yang et al. [33] integrated graph structures with Mamba and constructed a spatial–spectral cube to preserve spatial and spectral information.

Although SSM-based methods demonstrate advantages in long-range dependency modeling and computational efficiency, several challenges remain in HSI classification tasks. In some approaches, emphasis on sequence modeling efficiency leads to insufficient representation of fine-grained local features. In addition, directly applying a generic Mamba architecture to visual or hyperspectral tasks often fails to accommodate the data organization and modeling characteristics of spatial–spectral features. To address these limitations, a DPSF module is proposed by integrating visual characteristics with hyperspectral feature representation. This module enhances local feature representation while retaining the long-range dependency modeling advantages of the SSM.

III. Methods

A. Overview of the MVCH

The overall structure of MVCH is illustrated in Fig. 2. The entire process is divided into five main stages. The framework collaboratively models local spatial–spectral, structural relational, global contextual, local perceptual, and dynamic dependency views within a unified representation space, promoting complementary interaction and collaborative optimization among multiview features. First, HSIs $\bar{\mathcal{X}} \in \mathcal{R}^{\{C\times H\times W\}}$ are cropped into fixed size image patches $\mathcal{X} \in \mathcal{R}^{\{c\times h\times w\}}$ as input which preserves spatial context while reducing computational complexity (**Step 1**). Second, in **Step 2**, a preprocessing module is designed to perform a preliminary feature extraction on the input patches. This

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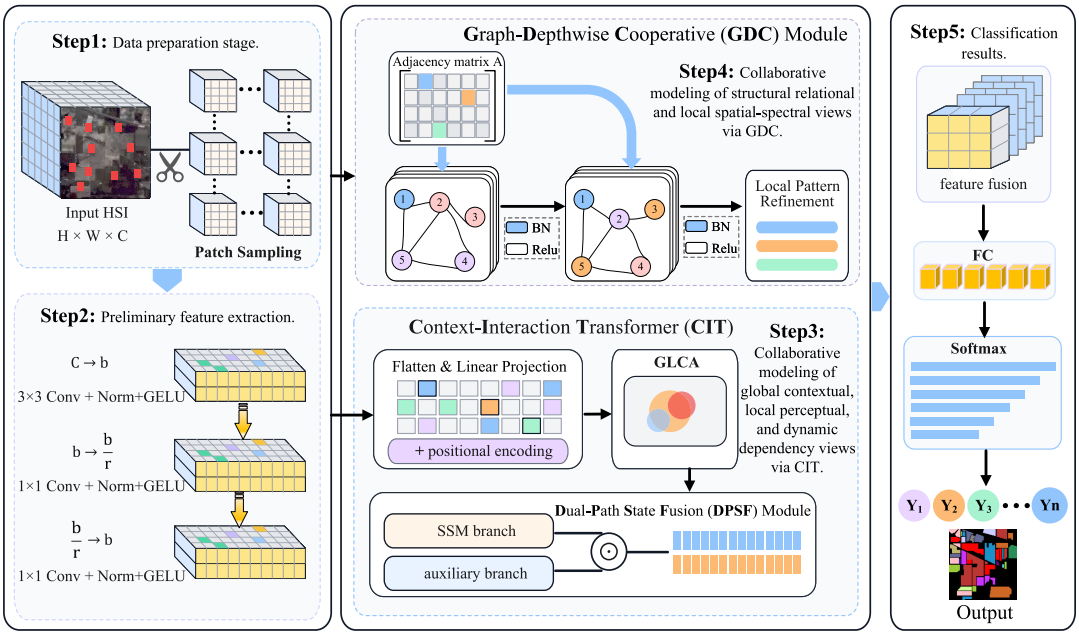


Fig. 2. Overall framework of MVCH. The framework performs multiview dependency modeling by integrating complementary representation views within a unified feature space. Specifically, GDC models structural relational and local spatial–spectral views, while CIT models global contextual, local perceptual, and dynamic dependency views through GLCA and DPSF. The complete pipeline consists of five stages: 1) HSI patch sampling; 2) preliminary spectral–spatial feature extraction; 3) contextual and dynamic dependency learning with CIT; 4) structural relational and local feature refinement with GDC; and 5) feature fusion and classification.

module integrates 2-D convolutional operations at multiple scales to suppress spatial feature redundancy while projecting high-dimensional spectral information into a low-dimensional feature space that is more suitable for discrimination. Through moderate dimensionality reduction and feature augmentation, this process effectively reduces parameter scale and computational overhead and provides a compact and discriminative spatial–spectral embedding representation for subsequent multiview feature modeling. The output of this module can be formulated as

$$Y_2 = C_1^{\frac{b}{\gamma}\to b}\left(C_1^{b\to\frac{b}{\gamma}}\left(C_3^{c\to b}(\mathcal{X})\right)\right) \tag{1}$$

where $C_3^{c\to b}(\cdot)$ denotes a convolution operation with a 3×3 kernel, $C_1(\cdot)$ denotes a convolution operation with a 1×1 kernel, and b is the spectral band.

In **Step 3**, Y_2 is fed into CIT, where the global contextual, local perceptual, and dynamic dependency views are collaboratively modeled to enhance contextual interaction and dynamic dependency representation. Subsequently, in **Step 4**, the feature Y_2 is fed into GDC to jointly model structural relational and local spatial–spectral views, enhancing the discriminative representation of fine-grained category differences and discontinuous structural regions. Finally, in **Step 5**, the complementary multiview features learned by CIT and GDC are fused in a unified representation space. The fused features are further refined by a convolutional layer and then fed into a fully connected layer for final classification. The overall process can be formulated as follows:

$$\hat{\mathcal{Y}} = Softmax\left(f\left(C_{2d}\left(GDC\left(Y_2\right)\right)\oplus CIT\left(Y_2\right)\right)\right) \tag{2}$$

where $GDC(\cdot)$ represents the operation of **Step 4**, $CIT(\cdot)$ represents the operation of **Step 3**, $C_{2d}(\cdot)$ represents the convolution operation of kernel, $\oplus$ represents feature fusion operation, $f(\cdot)$ represents feature learning through fully connected layers, and Y_{out} represents the prediction probabilities of different categories.

B. GDC Module

Graph-based methods effectively capture interpixel structural relationships through relational propagation, while local spatial–spectral modeling provides complementary fine-grained information. To coordinate these two types of dependency information, we propose GDC to jointly model structural relational and local spatial–spectral views within a unified framework.

Specifically, GDC first performs graph-based relational propagation to establish the structural relational view. Node connections are determined by feature similarity, and graph convolution aggregates correlated information to capture nonlocal structural dependencies. Depthwise separable convolution then refines the propagated features with local spatial–spectral information. Through the collaboration of these two views, GDC obtains more discriminative HSI representations. Formally, the GDC can be expressed as follows:

$$Y_{gdc} = D_c\left(G_2\left(G_1\left(Y_2\right)\right)\right) \tag{3}$$

where Y_2 denotes the input of the GDC, and $G(\cdot)$ and $D_c(\cdot)$ refer to the graph convolutional layer and the depthwise separable convolutional layer, respectively.

To improve computational efficiency and facilitate integration with CNN and ViT modules, a block-level graph modeling

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strategy is adopted. For the input image patch $\mathcal{X} \in \mathcal{R}^{\{c\times h\times w\}}$, each spatial location (h, w) is treated as a graph node, and its corresponding feature vector serves as the node attribute. Consequently, the image patch is transformed into a graph representation with a node feature matrix $\mathbf{h} \in \mathcal{R}^{N\times c}$, where $N = h \times w$ denotes the number of nodes.

During graph relational modeling, the connections between nodes are determined according to feature similarity. Specifically, the node features are first performed L2 normalization to eliminate the influence of feature magnitude. The similarity between any two nodes i and j is then computed using cosine similarity

$$S_{ij} = \mathbf{h}_i^T \mathbf{h}_j \tag{4}$$

where S_{ij}denotes the similarity score between nodes i and j. To reduce the influence of noisy connections and obtain a sparse graph structure, only the top-K most similar neighbors are retained for each node. Subsequently, the Softmax function is applied to normalize the retained similarity scores to obtain the sparse adjacency matrix $\mathbf{A}$

$$\mathbf{A}_{ij} = \frac{\exp\left(\frac{S_{ij}}{\tau}\right)}{\sum_{K\in\mathcal{N}_i}\exp\left(\frac{S_{iK}}{\tau}\right)} \tag{5}$$

where $\mathbf{A}_{ij}$ represents the element at the ith row and jth column of $\mathbf{A}$, which corresponds to the edge weight between nodes i and j. $\mathcal{N}_i$ denotes the top-K (K = 8) neighbor set of node i, and τ ($\tau = 0.2$) is a temperature parameter controlling the sharpness of the Softmax normalization. The parameter selection analysis of τ is presented in Section IV-E. To ensure a stable graph propagation process and satisfy the symmetry requirement of Chebyshev graph convolution, the adjacency matrix $\mathbf{A}$ is further symmetrized and augmented with self-loops

$$\tilde{\mathbf{A}} = \frac{\mathbf{A} + \mathbf{A}^T}{2} + \mathbf{I} \tag{6}$$

where $\mathbf{I}$ denotes the identity matrix. Based on the Chebyshev polynomial approximation, the graph convolution operation is defined in this work. For a sequential signal P [34], its $\mathcal{K}$th-order graph convolution can be expressed as

$$g_\theta \cdot P \approx \sum_{k=0}^{\mathcal{K}} \theta'_k T_k(\mathbf{L})\, P \tag{7}$$

where θ'_k denotes the Chebyshev coefficient, and the Chebyshev polynomial $T_k(\mathbf{L}) = 2\mathbf{L}T_{k-1}(\mathbf{L}) - T_{k-2}(\mathbf{L})$, with $T_0(\mathbf{L}) = \mathbf{I}$, $T_1(\mathbf{L}) = \mathbf{L}$. Here, g_θ represents the spectral filter, and $\mathbf{L}$ is the symmetrically normalized graph Laplacian matrix

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\tilde{\mathbf{A}}\mathbf{D}^{-\frac{1}{2}} \tag{8}$$

where $\mathbf{D}$ is the degree matrix of $\tilde{\mathbf{A}}$. By restricting $k = 1$ [35], the graph convolution can be further simplified to

$$g_\theta \cdot P \approx \theta\left(\mathbf{I} + \mathbf{D}^{-\frac{1}{2}}\tilde{\mathbf{A}}\mathbf{D}^{-\frac{1}{2}}\right) P. \tag{9}$$

Consequently, the propagation rule for GCN is defined as

$$\mathbf{H}^{(l)} = \delta\left(\mathbf{D}^{-\frac{1}{2}}\tilde{\mathbf{A}}\mathbf{D}^{-\frac{1}{2}}\mathbf{H}^{(l-1)}\mathbf{W}^{(l-1)} + \mathbf{b}^{(l-1)}\right) \tag{10}$$

where $\mathbf{H}^{(l)}$ denotes the output feature matrix of the lth layer, $\mathbf{W}^{(l-1)}$ and $\mathbf{b}^{(l-1)}$ represent the weight matrix and bias term, respectively, and δ is the ReLU activation function.

Since the designed GCN processes image patches of a specific size in each step, according to (10), the update rule for a batch can be directly expressed as

$$\mathbf{H}_m^{(l)} = \delta\left(\mathbf{D}_m^{-\frac{1}{2}}\tilde{\mathbf{A}}_m\mathbf{D}_m^{-\frac{1}{2}}\mathbf{H}_m^{(l-1)}\mathbf{W}_m^{(l-1)} + \mathbf{b}_m^{(l-1)}\right) \tag{11}$$

where m indicates the batch index during model training. After relational propagation, GDC further employs depthwise separable convolution for local spatial–spectral modeling, supplementing fine-grained neighborhood information with low computational cost and refining the graph-propagated features. The two modeling mechanisms interact collaboratively within the same representation space, enabling complementary modeling of the structural relational and local spatial–spectral views.

C. Context-Interaction Transformer

Unlike existing methods that focus predominantly on either global contextual modeling or sequential dependency learning, CIT integrates global context interaction, local perception enhancement, and dynamic dependency learning within a unified framework, enabling more comprehensive feature representation for HSI classification. As illustrated in Fig. 3(b), the feature map Y_2 is first serialized into a 1-D token sequence by flattening the spatial dimensions. The resulting tokens are then linearly projected and combined with learnable positional embeddings to generate the input representation T_{in} (see Step 3 in Fig. 2). This operation converts the spatial feature map into a unified token sequence format for subsequent sequence modeling

$$T_{in} = Projection\left(Flatten\left(Y_2\right)\right) + P_o \tag{12}$$

where Y_2 is the output of the preprocessing module, as defined in (1). P_o represents learnable position parameters.

First, T_{in} is normalized to stabilize the feature distribution. Afterward, T_{in} is fed into the GLCA. To better model spatial contextual interactions, GLCA temporarily reshapes the sequence into a 2-D spatial representation and performs global–local context aggregation in the spatial domain. After feature interaction, the output is converted back into the sequence format T_{GLCA}. It is worth noting that this sequence–spatial–sequence transformation is a lossless tensor rearrangement operation that does not change feature information. To alleviate the vanishing gradient problem in deep networks and retain the original feature information, we add a residual connection between T_{GLCA} and T_{in}, followed by another normalization operation to obtain the intermediate feature representation Y_{bn}

$$Y_{bn} = LN\left(GLCA\left(LN\left(T_{in}\right)\right) + T_{in}\right) \tag{13}$$

where $LN(\cdot)$ denotes the LayerNorm. $GLCA(\cdot)$ denotes the feature processing operations performed within the GLCA. Subsequently, Y_{bn} is delivered to DPSF for dynamic dependency modeling on contextual relations, preventing local details from being lost. Finally, the output feature of the DPSF

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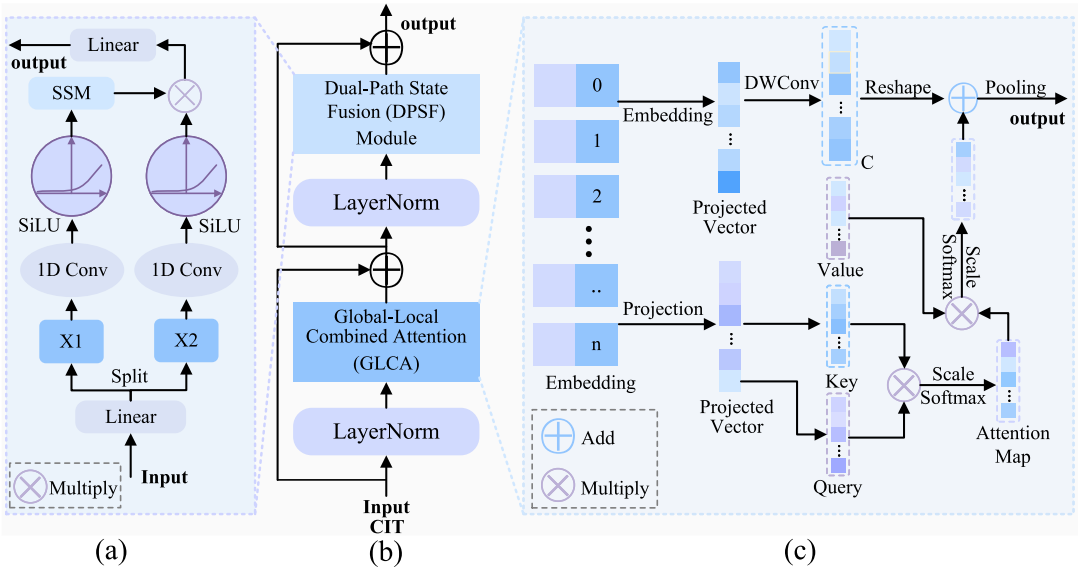


Fig. 3. (a) DPSF, which facilitates dynamic dependency modeling by capturing long-range dependencies while incorporating complementary spatial information. (b) CIT, which enhances contextual interaction and dynamic dependency representation by collaboratively modeling global contextual, local perceptual, and dynamic dependency views through GLCA and DPSF. (c) GLCA, which enhances contextual interaction by incorporating spatial-aware information into attention modeling to preserve local perception while capturing global dependencies.

is fused with Y_{bn} through a residual connection to obtain the final output feature Y_{cit} of the CIT module

$$Y_{cit} = DPSF(Y_{bn}) + Y_{bn} \tag{14}$$

where $DPSF(\cdot)$ denotes the feature processing operations performed within the DPSF.

1) Global–Local Combined Attention Mechanism: In HSI classification, the discrimination of ground objects depends not only on global distribution features, but also on local spatial structures and neighborhood consistency. However, pure ViT architectures often fail to capture local discriminative features, limiting performance in complex scenes. To overcome these limitations, a GLCA module is designed in Fig. 3(c). Specifically, given an input feature C_{in}, a convolutional projection is first applied to generate query, key, and value feature maps, expressed as

$$Q, K, V = Conv(C_{\text{in}}) \tag{15}$$

where $Conv(\cdot)$ denotes a convolution operation. Subsequently, similarity between Q and K is computed, normalized via the Softmax function, and multiplied by V

$$Scores = Softmax\left(\frac{QK^{\top}}{\sqrt{d_K}}\right)V \tag{16}$$

where d_K denotes the dimensionality of K. This process enables global modeling of dependencies across spatial positions and spectral channels, injecting global contextual information into the features. However, SA alone remains limited in capturing local spatial structures. To strengthen the representation of local features, GLCA introduces a convolutional embedding branch parallel to the attention branch. After BatchNorm and nonlinear transformation, local spatial information is extracted using depthwise separable convolution

$$S_{pa} = D_{wconv}\big(BN(G(C_{\text{in}}))\big) \tag{17}$$

where $G(\cdot)$ denotes the GELU activation function, and $D_{wconv}(\cdot)$ denotes depthwise separable convolution. This lightweight convolution operation significantly reduces parameter count and computational complexity while effectively enhancing modeling of local textures and structural details. By integrating both branches, the overall feature transformation of GLCA is defined as

$$T_{(Q,K,V,C)} = Softmax\left(\frac{QK^{\top}}{\sqrt{d_K}}\right)V + D_{wconv}\big(N(G(C))\big). \tag{18}$$

Subsequently, the output features T are mapped to the same dimension as the input features through a pooling operation, facilitating feature fusion and residual connections in subsequent modules. Within the proposed GLCA, multiple attention heads operate independently in parallel to learn diverse attention patterns. Therefore, outputs from all heads are concatenated to form the final GLCA output

$$T_{GLCA} = Concat\left(T^{(1)}, T^{(2)}, \ldots, T^{(h)}\right) \tag{19}$$

where h denotes the number of attention heads. Through this design, GLCA achieves collaborative integration of global contextual dependency modeling and local spatial structure perception. Not only alleviates limitations of SA in local feature modeling, but it also provides stable and discriminative feature representations for subsequent DPSF, improving HSI classification performance.

2) DPSF Module: Directly applying the generic SSM to visual feature modeling may weaken local discriminative details during dynamic dependency learning. To alleviate this issue, a DPSF module is proposed, as illustrated in Fig. 3(a). DPSF consists of an SSM branch for dynamic dependency modeling and an auxiliary branch without SSM for enhancing local feature representation during dependency learning. The outputs of the two branches are integrated

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through element-wise nonlinear feature modulation, allowing the model to jointly exploit long-range dependency information and complementary local feature representations. Furthermore, conventional 1-D convolution is adopted instead of causal convolution, further enhancing the perception of local feature variations while maintaining effective sequential context modeling.

Specifically, DPSF first performs channel expansion on the input feature representation D_{in} through a linear layer and then splits the expanded features into two symmetric paths

$$[\mathcal{X}_1, \mathcal{X}_2] = Split\,(Linear\,(D_{in}))\,. \tag{20}$$

In the SSM branch, 1-D depthwise separable convolution is first used for local operations. Afterward, the output is passed through a SiLU activation function and fed into an SSM to model long-range cross-token dependencies and dynamic evolution, obtaining feature representation D_1

$$\mathcal{D}_1 = SSM\,(\gamma\,(C_{1d}\,(\mathcal{X}_1))) \tag{21}$$

where $C_{1d}(\cdot)$ denotes a 1-D convolution layer and $\gamma(\cdot)$ represents the SiLU activation function, $\mathrm{SiLU}(x) = x/(1 + e^{-x})$. In the auxiliary branch, SSM is not introduced. The same local sequence processing and nonlinear mapping are performed to compensate for potential detail information loss that may arise from pure sequence modeling

$$\mathcal{D}_2 = \gamma\,(C_{1d}\,(\mathcal{X}_2))\,. \tag{22}$$

Subsequently, outputs from the two branches are fused through element-wise multiplication to achieve gated feature modulation. A linear projection is then applied to restore the original feature dimensionality and produce the final output

$$D_{out} = Linear\,(\mathcal{D}_1 \odot \mathcal{D}_2) \tag{23}$$

where $\odot$ denotes element-wise multiplication. Through this design, DPSF preserves long-range modeling capability provided by SSM while enhancing retention of local information via the auxiliary branch, enabling more stable and discriminative dynamic modeling of token sequences generated by GLCA. In this manner, DPSF and GLCA form a complementary functional relationship.

The SSM learns the hidden state $h(t) \in \mathcal{R}^M$ through learnable parameters $\mathcal{A} \in \mathcal{R}^{M\times M}$, $\mathcal{B} \in \mathcal{R}^{1\times M}$, $\mathcal{C} \in \mathcal{R}^{1\times M}$, and $\mathcal{D} \in \mathcal{R}^{M\times M}$, and transforms a 1-D continuous input $x(t) \in \mathcal{R}$ into $y(t) \in \mathcal{R}$. This process can be described by a linear ordinary differential equation with evolution parameters

$$\frac{dh\,(t)}{dt} = \mathcal{A}h\,(t) + \mathcal{B}x\,(t) \tag{24}$$

$$y\,(t) = \mathcal{C}h\,(t) + \mathcal{D}x\,(t) \tag{25}$$

where $x(t)$ denotes the input sequence, $h(t)$ represents the latent hidden state, and $y(t)$ corresponds to the predicted output. However, in deep learning applications, state transitions operate in discrete time rather than continuous time. Therefore, the above formulation can be converted into the following discrete form:

$$h_k = \bar{\mathcal{A}}h_{k-1} + \bar{\mathcal{B}}x_k \tag{26}$$

$$y_k = \bar{\mathcal{C}}h_k + \bar{\mathcal{D}}x_k \tag{27}$$

Algorithm 1 Algorithm Flow of MVCH

Input: HSI $\bar{\mathcal{X}} \in \mathcal{R}^{C\times H\times W}$; patch size s.
Output: Predicted label map $\mathcal{T}$.

```
Extract spatial-spectral patches X ∈ R^{c×s×s} centered at sampled pixels.
foreach epoch do
    Use the preprocessing module to obtain preliminary spatial-spectral features : Y_2 ⟵ X;
    GDC branch: collaborative modeling of structural relational and local spatial-spectral views: Y_gdc ⟵ GDC(Y_2);
    CIT branch: collaborative modeling of global contextual, local perceptual, and dynamic dependency views: Y_cit ⟵ DPSF(GLCA(Y_2));
    Fuse features from the two branches: Y^fusion = C_2d(Y_gdc ⊕ Y_cit);
    Predict class probabilities : Ŷ = Softmax(f(Y^fusion)).
end
Predicted label map T.
```

where $\bar{\mathcal{A}} = e^{\Delta\mathcal{A}}$, $\bar{\mathcal{B}} = (e^{\Delta\mathcal{A}} - I)\mathcal{A}^{-1}\mathcal{B}$, $\bar{\mathcal{C}} = \mathcal{C}$, Δ represents the time step, h_{k-1}, and h_k denote the hidden states at the previous and current time steps, respectively. The overall training details are shown in Algorithm 1.

IV. Experiments and Analysis

A. Dataset Description

Three widely used HSI benchmark datasets are selected for comparative experiments to evaluate the performance of the algorithm proposed in this article: Salinas Valley (SV), Pavia University (PU), and IndianPines (IP). The details of the datasets are shown in Table I.

1) SV Dataset [36]: The SV dataset represents a classical application of HSI technology in agricultural remote sensing analysis. The dataset consists of 204 spectral bands with a spatial resolution of 512×217 pixels. The scene mainly covers agricultural areas in the SV, including crops such as fallow, celery, and stubble, and contains 16 ground-truth classes. For experimental evaluation, 0.8% of the samples from each class are randomly selected for training, while the remaining samples are used for testing.

2) IP Dataset [37]: The IP dataset is a widely used benchmark for HSI classification in agricultural remote sensing. The dataset consists of 200 spectral bands with a spatial resolution of 145×145 pixels. The dataset includes 16 ground-truth classes, such as corn, soybean, and grass. In the experiments, 0.8% of the samples from each class are used for training, and the rest are reserved for testing.

3) PU Dataset [38]: The PU dataset is a representative benchmark for HSI analysis in urban scenes. The dataset contains 103 spectral bands with a spatial size of 610×340 pixels and includes 9 ground-truth classes, such as asphalt, meadows, and gravel. In the experimental setup, 0.8% of the samples from each class are selected for training, and the remaining samples are used for testing.

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR SV, IP, AND PU DATASETS

No.	SV				IP				PU			
	Class name	Total	Train	Test	Class name	Total	Train	Test	Class name	Total	Train	Test
1	Broccoli_weeds_1	2009	16	1993	Alfalfa	46	1	45	Asphalt	6631	53	6578
2	Broccoli_weeds_2	3726	29	3697	Corn_notill	1428	11	1417	Meadows	18649	149	18500
3	Fallow	1976	15	1961	Corn_mintill	830	6	824	Gravel	2099	16	2083
4	Fallow_rough_plow	1394	11	1383	Corn	237	1	236	Trees	3064	24	3040
5	Fallow_smooth	2678	21	2657	Grass_pasture	483	3	480	Metal_sheets	1345	10	1335
6	Stubble	3959	31	3928	Grass_trees	730	5	725	Bare_soil	5029	40	4989
7	Celery	3579	28	3551	Grass_pasture_mowed	28	1	27	Bitumen	1330	10	1320
8	Grapes_untrained	11271	90	11181	Hay_windrowed	478	3	475	Bricks	3682	29	3653
9	Vinyard_develop	6203	49	6154	Oats	20	1	19	Shadows	947	7	940
10	Green_weeds	3278	26	3252	Soybean_notill	972	7	965	–	–	–	–
11	Lettuce_4wk	1068	8	1060	Soybean_mintill	2455	19	2436	–	–	–	–
12	Lettuce_5wk	1927	15	1912	Soybean_clean	593	4	589	–	–	–	–
13	Lettuce_6wk	916	7	909	Wheat	205	1	204	–	–	–	–
14	Lettuce_7wk	1070	8	1062	Woods	1265	10	1255	–	–	–	–
15	Vinyard_untrained	7268	58	7210	Buildings-Grass-Trees-Drives	386	3	383	–	–	–	–
16	Vinyard_trellis	1807	14	1793	Stone-Steel-Towers	93	1	92	–	–	–	–
Total	–	54129	426	53703	–	10249	77	10172	–	42776	338	42438

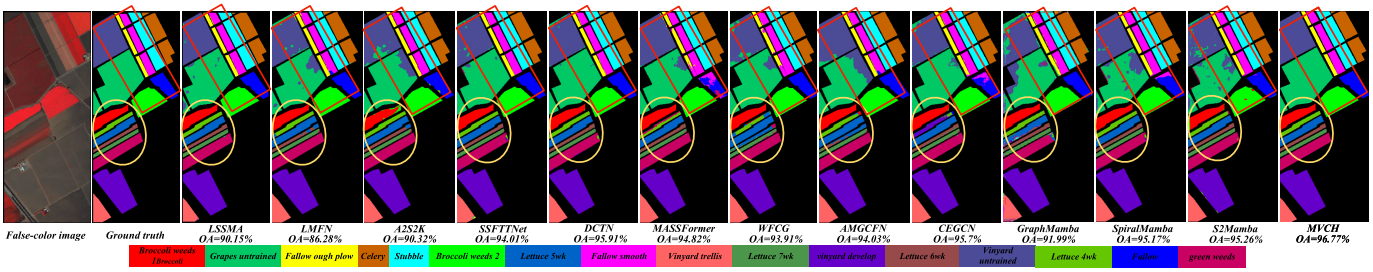


Fig. 4. Classification maps obtained by different methods on the SV dataset.

B. Experimental Settings

1) Environment Setup: In our experiments, we repeated each experiment five times and reported the average result and standard deviation of each metric to avoid the effect of random operations. The MVCH network proposed in this article and the other comparison networks are implemented on the PyTorch platform. All networks were run on a system equipped with an NVIDIA RTX 5000 Ada GPU and 32 GB of RAM.

2) Parameter Setup: We ran the code for 12 classification methods published online and attempted to achieve the highest classification accuracy in the three hyperspectral datasets with the same hyperparameter settings for all the methods. Adam optimizer is used to optimize the network parameters with a batch size of 100. The number of epochs is set to 200 based on a series of studies. The learning rate and patch size were set to 5e-4 and 11×11, respectively.

3) Comparison Methods: HSI data involves complex spatial structures, high-dimensional spectral information, and intricate pixel relationships. Different methods typically emphasize different aspects of hyperspectral representation learning. To comprehensively evaluate the effectiveness of the proposed MVCH, representative methods from four categories are selected for comparison. Specifically, three CNN-based methods, LSSMA [39], LMFN [40], and A2S2K [41], are adopted to represent local spatial–spectral feature learning. Three Transformer-based methods, including SSFTTNet [42], DCTN [43], and MASSFormer [44], are selected to represent global dependency modeling. Three GCN-based methods, WFCG [45], AMGCFN [46], and CEGCN [47], are employed to characterize graph-based relational reasoning. In addition, three SSM-based methods, GraphMamba [33], SpiralMamba [48], and S2Mamba [49], are included to represent the emerging state-space sequence modeling approach. Collectively, these methods cover diverse perspectives of hyperspectral representation learning and provide a comprehensive benchmark for evaluating the proposed MVCH.

C. Comparison With Representative Methods

1) SV Dataset: As shown in Table II, achieves the highest OA, AA, and Kappa values among the compared methods, indicating its effectiveness in classification performance. The yellow circled region in Fig. 4 contains four adjacent lettuce categories, namely Lettuce 4wk–7wk (Classes 11–14), which correspond to different growth stages of the same crop. Since differences in growth stages lead to variations in local spatial structures and spectral characteristics, these categories are highly prone to confusion. Methods such as LMFN, WFCG, CEGCN, and GraphMamba exhibit obvious misclassifications in this region. In contrast, MVCH can more accurately distinguish different growth stages, indicating its effectiveness in capturing fine-grained spatial–spectral differences. The red boxed region contains different categories that

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TABLE II

COMPARISON OF CLASSIFICATION PERFORMANCE (AA, OA, AND K) ON THE SV DATASET. (BOLD INDICATES OUR RESULTS)

No.	CNN-based			Transformer-based			GCN-based			SSM-based			Ours
	LSSMA	LMFN	A2S2K	SSFTTNet	DCTN	MASSFormer	WFCG	AMGCFN	CEGCN	GraphMamba	SpiralMamba	S2Mamba	MVCH
1	94.88±0.86	87.59±9.66	94.33±1.53	95.48±6.81	94.89±1.34	99.97±0.07	98.09±0.81	96.32±4.7	99.90±0.23	99.22±1.28	98.53±2.08	91.26±1.07	**96.41±1.32**
2	99.44±0.45	95.72±4.30	99.30±0.61	100.00±0.00	100.00±0.00	99.96±0.06	99.43±0.69	99.99±1.08	100.00±0.00	100.00±0.00	99.75±0.57	99.12±0.84	**100.00±0.00**
3	94.38±3.67	93.43±2.27	79.33±2.90	96.80±6.54	99.50±0.23	99.99±0.02	95.51±9.10	99.44±8.89	66.90±26.63	99.56±0.32	95.50±5.89	93.97±3.08	**98.89±1.79**
4	97.00±0.75	93.10±4.45	97.25±0.38	90.65±8.71	96.66±3.23	98.87±1.30	99.54±1.02	98.92±1.53	94.09±8.64	93.95±2.45	97.45±1.47	97.57±1.81	**98.96±0.25**
5	96.72±2.10	94.45±2.30	90.53±0.62	95.99±8.31	97.90±2.07	96.19±1.46	82.29±8.15	94.42±2.78	99.16±0.59	97.73±0.82	99.69±0.40	98.90±1.00	**99.61±0.24**
6	98.38±0.04	97.91±0.52	97.56±0.22	99.77±0.33	97.90±0.72	99.89±0.22	99.78±0.40	99.38±8.87	99.82±0.28	98.13±0.79	99.99±0.01	99.83±0.35	**98.43±0.42**
7	98.44±0.32	94.55±1.82	97.39±0.47	99.48±0.45	97.58±1.24	99.97±0.03	99.59±0.43	99.92±1.19	99.98±0.03	98.26±0.35	99.93±0.14	98.46±0.29	**95.66±7.08**
8	86.92±1.68	83.63±2.51	92.27±2.45	83.89±6.49	98.18±0.58	89.67±0.91	95.73±1.89	87.16±3.01	94.63±1.81	86.20±3.39	89.10±3.86	89.21±1.69	**99.10±0.28**
9	99.34±0.13	96.97±1.03	96.95±2.20	99.89±0.22	99.66±0.19	99.11±0.26	97.05±4.31	99.99±1.3	100.00±0.00	99.83±0.17	100.00±0.01	99.28±0.36	**99.43±0.31**
10	91.58±2.56	88.09±3.30	92.29±3.12	96.40±2.14	96.97±0.91	96.10±1.91	98.56±0.99	99.26±7.64	98.57±1.20	87.84±3.07	97.75±2.17	98.60±0.78	**98.50±0.58**
11	95.74±0.55	82.27±15.20	91.39±4.02	98.44±1.16	96.00±1.37	99.47±1.14	100.00±0.00	90.5±2.88	63.22±35.92	95.59±3.60	90.23±7.13	93.70±2.20	**96.33±1.23**
12	97.28±0.22	95.45±1.52	90.72±5.83	94.90±4.52	96.86±0.30	99.64±0.27	75.99±14.11	99.85±2.94	99.97±0.07	80.86±7.08	98.80±0.85	96.52±1.35	**97.24±0.55**
13	95.06±4.33	94.54±1.73	92.41±3.05	99.32±1.17	96.82±0.91	94.29±5.56	98.86±2.54	59.14±3.68	67.12±32.86	89.84±4.30	95.68±4.76	97.85±2.13	**98.63±0.88**
14	94.24±4.40	92.12±2.15	95.21±2.19	94.83±5.99	95.55±1.95	88.30±6.50	84.59±16.31	97.67±6.2	96.78±1.32	97.02±3.00	98.80±0.85	97.13±1.18	**98.23±1.03**
15	75.24±4.03	72.81±3.10	89.01±2.29	91.12±5.30	95.71±0.79	85.23±1.55	92.70±2.35	91.05±5.17	98.04±1.68	79.72±2.78	85.32±13.07	90.30±3.12	**96.33±0.12**
16	84.54±1.30	76.94±5.30	65.72±19.24	99.15±0.43	86.25±0.42	99.68±0.62	99.99±0.03	100.00±0.00	98.46±2.64	87.38±7.23	99.83±0.17	99.66±0.56	**83.91±10.31**
AA(%)	93.70±0.86	89.97±2.43	91.35±0.92	96.01±0.54	96.65±0.33	96.26±0.71	94.86±0.91	94.56±2.4	92.34±1.63	93.20±0.67	96.65±0.39	96.33±0.47	**97.23±1.34**
OA(%)	90.04±0.82	86.89±1.67	90.44±1.00	94.09±0.83	96.06±0.13	94.92±0.40	94.45±1.29	94.63±0.92	95.71±1.06	91.66±0.37	94.97±1.11	95.09±0.31	**96.63±0.45**
K×100	88.94±0.88	85.45±1.88	89.10±1.14	93.44±0.92	95.63±0.15	94.34±0.45	93.81±1.45	94.03±1.02	95.23±1.18	90.72±0.41	94.40±1.25	94.54±0.35	**96.15±0.47**

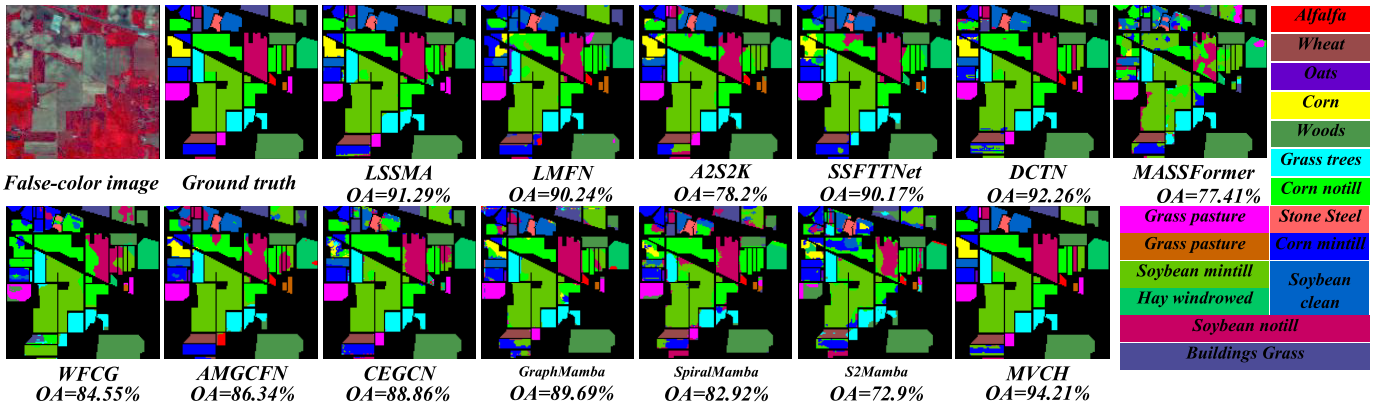


Fig. 5. Classification maps obtained by different methods on the IP dataset.

are spatially adjacent and have complex boundaries. CEGCN, MASSFormer, and S2Mamba produce a large number of isolated misclassified pixels and fragmented patches in this region, causing blurred class boundaries. Compared with these methods, the regions generated by MVCH are more complete and exhibit clearer boundaries, enabling more effective preservation of spatial structural continuity and reduction of interclass confusion.

2) IP Dataset: As shown in Table III, MVCH improves OA by 1.85% and Kappa by 2.13% compared with the second-best method DCTN. Moreover, the OA gain reaches 22.47% over S2Mamba. The IP dataset contains multiple agricultural categories with highly similar spectral signatures and subtle interclass differences caused by different cultivation conditions. Therefore, accurate classification requires not only contextual modeling but also the preservation of fine-grained local spatial–spectral discriminative information. Although methods such as LSSMA, DCTN, and GraphMamba achieve competitive performance by exploiting local spatial–spectral features, contextual dependencies, or relational information, they primarily emphasize specific representation perspectives. In contrast, MVCH adopts a multiview collaborative representation strategy that integrates the complementary strengths of different representation perspectives, enabling more discriminative feature learning and more effective discrimination of highly similar categories. Furthermore, as shown in Fig. 5, MVCH not only achieves high classification accuracy but also generates classification maps that are highly consistent with the ground truth.

3) PU Dataset: Compared with the IP dataset, the PU dataset exhibits stronger spatial continuity and more regular regional structures. Therefore, most methods are able to achieve relatively high classification accuracy. As shown in Table IV, the OA values of AMGCFN, MASSFormer, CEGCN, SpiralMamba, and S2Mamba all exceed 95%, indicating that existing methods can effectively exploit regional structural information. Nevertheless, MVCH still achieves the highest OA (96.29%), AA (97.57%), and Kappa (95.08%). As illustrated in Fig. 6, although MASSFormer, SpiralMamba, and S2Mamba achieve competitive quantitative results, bricks are incorrectly classified as gravel, causing the disruption of structural continuity within the target regions. CEGCN misclassifies bitumen as asphalt, while the SSM-based methods also exhibit varying degrees of category confusion in

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TABLE III

COMPARISON OF CLASSIFICATION PERFORMANCE (AA, OA, AND K) ON THE IP DATASET. (BOLD INDICATES OUR RESULTS)

No.	CNN-based			Transformer-based			GCN-based			SSM-based			Ours
	LSSMA	LMFN	A2S2K	SSFTTNet	DCTN	MASSFormer	WFCG	AMGCFN	CEGCN	GraphMamba	SpiralMamba	S2Mamba	MVCH
1	97.10±2.69	96.76±4.83	59.69±20.43	90.79±20.59	99.78±0.49	0.00±0.00	100.00±0.00	95.98±2.18	7.43±7.22	97.67±2.85	99.53±1.04	89.77±7.82	**99.78±0.48**
2	98.43±1.19	97.21±0.77	83.51±8.73	96.92±1.81	97.27±2.24	67.44±2.19	78.72±5.56	78.92±4.12	91.09±5.82	83.76±2.35	73.96±9.53	55.00±3.75	**99.62±0.34**
3	85.42±6.31	85.00±0.99	48.76±6.86	80.05±3.62	78.29±8.13	74.45±2.29	71.55±11.91	86.04±9.35	87.79±6.39	80.64±2.80	60.36±13.40	72.02±12.80	**89.62±0.98**
4	93.84±1.96	93.29±0.70	71.44±21.00	92.83±2.61	94.93±1.44	73.38±8.84	87.48±11.93	52.52±17.91	45.74±27.04	69.70±15.58	74.87±12.39	51.45±12.64	**96.47±0.09**
5	89.15±2.72	89.50±0.90	75.93±4.91	87.72±3.37	93.06±0.47	99.65±0.63	92.61±6.85	72.87±7.83	89.82±4.53	88.90±1.83	70.83±11.96	65.33±88.23	**93.36±0.05**
6	99.25±0.55	98.58±0.47	91.00±3.77	99.32±0.47	99.70±0.45	92.96±3.65	92.17±4.52	94.01±3.44	98.78±0.99	98.32±0.33	92.28±4.73	93.01±4.16	**99.88±0.17**
7	75.76±24.04	97.76±2.45	65.41±28.95	77.69±43.62	98.50±2.46	0.00±0.00	0.00±0.00	100.00±0.00	23.17±25.81	100.00±0.00	96.00±6.93	100.00±0.00	**99.65±0.78**
8	94.72±0.10	94.46±0.57	89.94±1.82	95.75±2.29	100.00±0.00	84.12±3.82	92.87±3.99	98.64±1.68	99.96±0.10	98.17±2.36	78.02±16.25	93.34±1.42	**97.42±0.00**
9	79.60±28.01	98.00±2.75	54.07±35.81	97.38±2.63	99.49±1.14	0.00±0.00	0.00±0.00	100.00±0.00	13.37±10.67	87.06±12.75	100.00±0.00	97.65±5.26	**99.49±1.14**
10	96.66±3.77	93.32±1.12	86.77±2.84	93.82±0.66	93.85±2.20	75.82±3.82	78.89±11.12	82.58±4.43	87.01±7.67	85.73±1.62	82.97±6.60	60.51±7.88	**95.14±1.13**
11	95.99±2.07	96.17±0.45	87.13±2.11	95.94±0.48	93.93±1.94	70.42±1.54	88.15±3.42	92.6±3.12	94.54±2.59	95.88±1.06	83.80±6.65	80.65±2.51	**97.34±0.28**
12	94.63±2.37	93.15±1.22	73.71±12.51	91.71±5.56	94.88±2.23	75.82±4.51	64.51±7.45	72.94±14.36	72.37±20.57	92.89±2.89	71.82±8.36	34.85±7.91	**96.99±0.14**
13	99.47±0.77	93.21±10.24	40.32±27.32	91.46±6.22	99.35±1.33	97.40±4.08	100.00±0.00	98.19±2.43	99.69±0.46	98.01±0.99	98.12±3.22	96.44±3.61	**98.07±4.31**
14	98.86±1.42	97.56±0.67	92.06±3.15	98.91±0.99	99.71±0.53	86.03±0.81	98.00±1.61	92.97±3.98	99.70±0.24	98.61±0.40	96.72±2.34	97.46±1.57	**99.89±0.11**
15	73.51±13.43	71.91±3.69	49.05±24.64	73.34±2.76	79.76±2.44	73.92±5.90	79.71±13.89	83.69±9.28	76.91±13.61	90.71±4.27	80.68±3.61	50.18±8.37	**80.82±0.31**
16	93.99±3.02	96.27±2.56	56.81±39.23	95.91±3.30	99.47±0.37	83.82±6.90	100.00±0.00	80.85±19.88	85.40±9.52	72.89±15.52	98.44±3.48	99.56±0.61	**99.25±0.91**
AA(%)	91.65±4.52	93.26±0.61	70.35±3.01	91.22±4.17	95.12±1.14	59.65±2.38	76.54±1.19	86.43±2.18	73.29±5.96	89.94±1.17	84.98±0.59	74.83±0.41	**96.42±0.44**
OA(%)	91.91±3.04	90.63±0.22	78.33±2.53	90.29±1.63	92.14±1.53	77.49±0.30	84.36±1.06	86.37±1.32	89.96±3.53	91.15±0.95	81.38±1.25	71.52±0.78	**93.99±0.35**
K×100	90.86±3.42	89.40±0.24	75.39±2.79	89.01±1.83	91.06±1.75	73.96±0.33	82.09±1.23	84.5±1.48	88.47±4.09	88.67±3.40	78.83±1.45	67.39±0.97	**93.19±0.40**

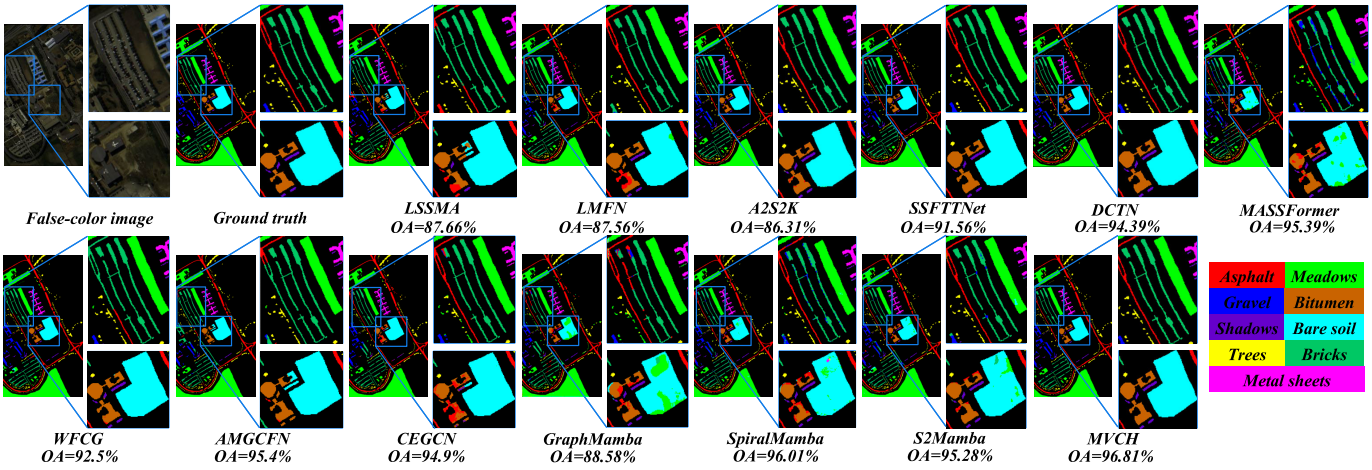


Fig. 6. Classification maps obtained by different methods on the PU dataset.

TABLE IV

COMPARISON OF CLASSIFICATION PERFORMANCE (AA, OA, AND K) ON THE PU DATASET. (BOLD INDICATES OUR RESULTS)

No.	CNN-based			Transformer-based			GCN-based			SSM-based			Ours
	LSSMA	LMFN	A2S2K	SSFTTNet	DCTN	MASSFormer	WFCG	AMGCFN	CEGCN	GraphMamba	SpiralMamba	S2Mamba	MVCH
1	96.62±0.35	96.52±0.32	91.83±0.92	94.05±2.04	97.63±0.11	95.34±0.83	85.01±6.34	90.79±4.61	99.39±0.77	94.56±1.01	97.04±1.66	92.99±1.48	**97.13±1.08**
2	90.74±1.38	90.53±1.38	89.49±2.97	91.36±1.86	94.81±0.02	97.47±0.99	98.48±1.33	99.81±0.14	99.9±0.18	97.66±1.50	98.76±0.61	98.42±0.21	**99.44±0.45**
3	91.63±4.28	91.21±4.08	74.82±3.89	83.92±0.95	95.50±0.44	86.73±7.18	92.76±2.42	83.92±18.36	77.96±8.72	57.77±3.33	69.56±20.90	80.27±5.50	**94.17±2.60**
4	95.68±0.49	95.36±0.63	91.88±1.47	96.81±1.11	98.59±0.04	96.87±1.18	96.99±2.26	87.7±4.95	78.91±5.69	61.26±7.51	92.89±3.12	95.81±0.63	**98.19±0.94**
5	99.55±0.26	99.50±0.22	98.37±1.03	99.90±0.07	100.00±0.00	97.94±1.28	86.96±5.78	99.94±0.05	100.00±0.00	99.97±0.07	99.22±0.54	98.74±1.04	**99.72±0.56**
6	99.80±0.20	99.74±0.15	94.70±2.55	99.97±0.05	96.49±0.34	97.69±1.73	98.20±0.98	99.93±0.14	99.09±0.85	81.80±6.06	93.77±4.15	93.14±2.55	**96.61±3.75**
7	99.90±0.15	99.96±0.08	87.67±4.15	99.83±0.39	74.92±8.18	93.59±4.10	84.03±7.52	90.4±8.38	66.57±35.5	78.52±11.68	83.27±11.14	87.03±3.52	**100.00±0.00**
8	98.18±1.35	98.17±1.34	91.62±1.70	99.30±1.05	99.77±0.09	87.13±2.89	78.98±10.24	98.24±0.68	98.11±1.38	86.82±5.43	94.23±5.02	95.81±0.86	**95.24±1.77**
9	97.06±0.96	96.94±0.93	90.04±2.47	99.33±0.16	97.62±2.50	97.42±1.31	100.00±0.00	66.47±14.99	67±5.33	73.64±11.75	97.65±1.29	96.35±1.23	**97.62±2.50**
AA(%)	96.58±0.76	96.44±0.70	90.05±1.21	96.05±0.55	95.04±0.92	92.31±1.41	91.27±2.63	90.8±3.67	87.44±4.85	81.34±2.17	91.82±3.08	93.17±0.55	**97.57±0.60**
OA(%)	89.81±1.10	89.68±1.04	86.21±2.67	91.52±1.76	94.38±0.04	95.60±0.91	92.85±2.69	95.61±1.39	95.23±1.73	88.76±0.78	95.18±1.08	95.26±0.31	**96.29±0.54**
K×100	86.47±1.42	86.38±1.38	81.65±3.53	88.97±2.38	92.75±0.05	94.14±1.22	90.43±3.61	94.18±1.84	93.63±2.3	84.86±1.01	93.58±1.47	93.72±0.41	**95.08±0.71**

the bare soil region. Although these misclassifications have a limited impact on the overall accuracy, they reduce the consistency between the classification maps and the actual spatial distribution of ground objects. In contrast, MVCH effectively preserves regional structures while reducing isolated misclassifications, leading to more spatially coherent classification results.

D. Ablation Study

1) Ablation Study of Different Branches in MVCH: To comprehensively evaluate the effectiveness of the proposed approach, ablation experiments are conducted on the branches containing GDC and CIT, with the results summarized in Table V. It can be observed that the complete MVCH consistently achieves the highest classification accuracy across all

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TABLE V
COMPARISON OF OA% FOR DIFFERENT BRANCH ABLATION STUDIES ON THE SV, IP, AND PU DATASETS

Methods	Datasets		
	SV	IP	PU
Full model	96.63 ± 0.45	93.99 ± 0.35	96.29 ± 0.54
Without GDC branch	94.11 ± 1.44	92.91 ± 2.80	91.49 ± 0.48
Without CIT branch	93.41 ± 0.68	92.59 ± 0.40	90.97 ± 1.61

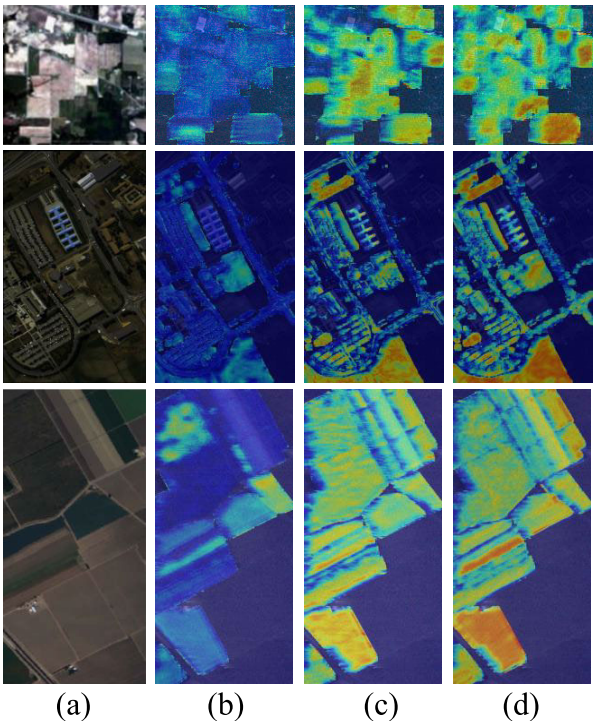


Fig. 7. Grad-CAM [50] visualizations at different layers on different datasets. (a) Original image. (b) GDC layer. (c) CIT layer. (d) Final layer. Colors from cool to warm indicate low to high contributions to target class discrimination. GDC mainly focuses on local fine-grained responses, and CIT emphasizes global region-level activations. The final layer integrates both characteristics, resulting in a balanced and discriminative activation distribution.

three datasets while exhibiting improved stability. The removal of either branch leads to varying degrees of performance degradation, indicating a clear complementary relationship between GDC and CIT in feature representation. In particular, the exclusion of CIT generally results in more pronounced accuracy degradation, highlighting the critical role of global pattern learning in HSI classification. Moreover, the absence of GDC weakens the capability to characterize local structures and fine-grained variations. On the PU dataset, the complete model attains an OA of 96.29%, whereas the removal of GDC or CIT causes substantial accuracy drops to 91.49% and 90.97%, respectively. These results demonstrate that, in scenarios characterized by complex spatial structures and pronounced spatial–spectral heterogeneity, relying solely on either local or global modeling strategies is insufficient, and the coordinated integration of GDC and CIT is essential for accurately capturing the spatial distribution of complex ground objects.

Fig. 7 shows the spatial response of MVCH at different layers. The activations of the GDC layer are generally sparse and locally concentrated, with strong responses mainly appearing at object boundaries and regions with pronounced local variations, indicating that GDC is more effective for discriminating complex boundaries and small-scale heterogeneous areas. In contrast, the CIT layer shows more continuous and region-level high responses over large areas, demonstrating its ability to model long-range contextual information and nonlinear spatial–spectral dependencies from a global perspective, providing stable discriminative evidence for homogeneous land-cover regions. Furthermore, the activations of the final layer of MVCH exhibit both global coherence and local selectivity while maintaining overall attention to the main land-cover regions, enhancing critical local details, and suppressing the diffusion of irrelevant background responses.

TABLE VI
COMPARISON OF OA% FOR DIFFERENT MODULES IN THE CIT ABLATION STUDY ON THE SV, IP, AND PU DATASETS

Methods	Datasets		
	SV	IP	PU
SA + MLP	94.88 ± 0.30	91.08 ± 0.90	91.48 ± 1.27
GLCA + MLP	95.86 ± 0.32	92.21 ± 0.72	91.33 ± 1.27
SA + DPSF	95.23 ± 1.20	91.55 ± 0.34	91.95 ± 1.03
Proposed CIT	96.63 ± 0.45	93.99 ± 0.35	96.29 ± 0.54

2) Ablation Study of Different Modules in CIT: To validate the effectiveness of the proposed components in the CIT, ablation experiments are conducted on the SV, IP, and PU datasets. A traditional Transformer consisting of SA and MLP is adopted as the baseline [10]. To fairly evaluate the contribution of each component, GLCA and DPSF are separately introduced by replacing SA and MLP, respectively, while keeping the remaining architecture unchanged.

As shown in Table VI, the traditional Transformer baseline (SA + MLP) achieves competitive classification performance on all datasets. Replacing SA with GLCA (GLCA + MLP) or substituting MLP with DPSF (SA + DPSF) further improves the performance, indicating that both components can effectively enhance hyperspectral feature representation. When GLCA and DPSF are simultaneously integrated, the proposed CIT (GLCA + DPSF) achieves the best results on all three datasets, demonstrating the complementarity between the two components. Further analysis reveals that the performance improvement brought by GLCA is more pronounced on the SV and IP datasets, suggesting that introducing a local enhancement mechanism into the attention framework is beneficial for improving feature discriminability in scenarios characterized by fine-grained spatial–spectral differences among categories. In contrast, DPSF contributes more significantly on the PU dataset, indicating that local structural refinement and dynamic dependency modeling of feature sequences can further improve feature representation in scenes with stronger spatial continuity and more regular regional structures.

As shown in Fig. 8, Row 1 presents spatially adjacent features with highly similar spectral characteristics that belong to the same class but correspond to different temporal cycles. The proposed CIT achieves relatively complete classification for such features, whereas the ablation methods exhibit varying

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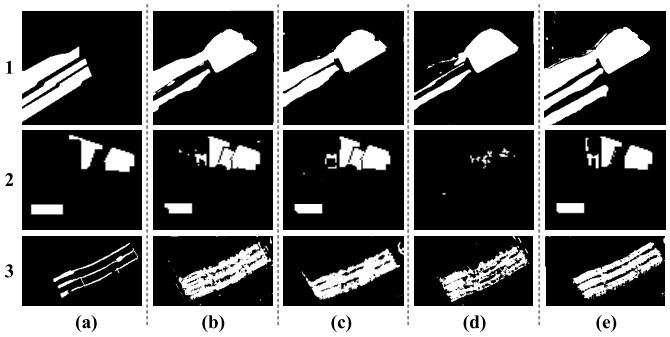


Fig. 8. Visualization of ablation on different modules in CIT. (a) Ground truth. (b) SA + DPSF. (c) GLCA + MLP. (d) SA + MLP. (e) Proposed CIT.

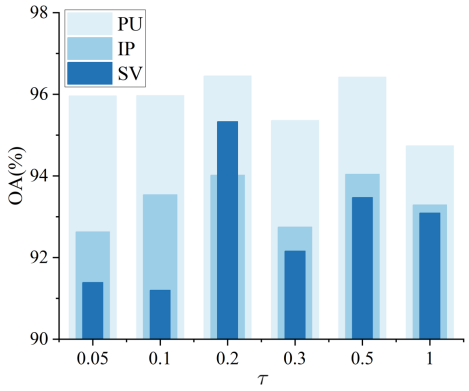


Fig. 9. OA (%) under different temperature parameter settings τ.

degrees of misclassification. Row 2 corresponds to objects with relatively regular spatial distributions across different regions. The results indicate that the configurations shown in Fig. 8(b) and (c) tend to misclassify adjacent objects, while the model in Fig. 8(d) fails to identify the target objects effectively. In contrast, the proposed CIT maintains clearer spatial boundaries and higher category consistency, demonstrating that the incorporation of local perception and structured sequence modeling effectively suppresses spatial neighborhood interference. Row 3 presents features characterized by continuity and linear structures. Although all methods capture the overall distribution trends, the ablation models generally suffer from excessive smoothing and blurred boundaries. By comparison, the complete CIT demonstrates better performance in preserving structural continuity and boundary integrity, validating its advantage in modeling continuous features.

E. Analysis of Different Temperature Parameter Settings τ

The temperature parameter τ controls the sharpness of edge-weight distributions during similarity normalization in the dynamic graph construction process. A smaller τ increases the distinction among similarity scores and produces more concentrated edge weights, while a larger τ generates smoother edge-weight distributions. Therefore, an appropriate τ is required to balance selective neighbor aggregation and effective structural information propagation.

To investigate the influence of τ on the proposed GDC module, different temperature parameters ranging from 0.05 to 1 are evaluated on three benchmark HSI datasets. As shown in Fig. 9, the proposed method achieves the best overall performance with $\tau = 0.2$. Although the classification accuracy varies under different τ settings, the model maintains relatively stable performance across the tested range, demonstrating the robustness of the proposed dynamic graph construction under different temperature parameter settings. Therefore, $\tau = 0.2$ is adopted in all experiments.

F. Feature Discriminability Analysis of DPSF

To quantitatively evaluate the discriminability of features from different branches, the Fisher ratio (FR) [51], based on the Fisher discriminant criterion, is adopted as the evaluation metric. It is defined as the ratio between the trace of the interclass scatter matrix and the intraclass scatter matrix

$$FR = \frac{tr(S_B)}{tr(S_W)} \tag{28}$$

where S_B and S_W denote the interclass and intraclass scatter matrices, respectively. Specifically, they are calculated as

$$S_B = \sum_{c=1}^{C} N_c (\mu_c - \mu)(\mu_c - \mu)^T \tag{29}$$

$$S_W = \sum_{c=1}^{C} \sum_{x_i \in c} (x_i - \mu_c)(x_i - \mu_c)^T \tag{30}$$

where N_c represents the number of samples in class c, μ_c denotes the mean feature of class c, and μ denotes the global mean feature, x_i represents the feature of the ith sample in class c. A larger FR indicates stronger class separability and more discriminative feature representations.

TABLE VII

QUANTITATIVE ANALYSIS OF FEATURE DISCRIMINABILITY FOR DIFFERENT DPSF COMPONENTS USING FR. THE SSM BRANCH DENOTES THE SSM-BASED BRANCH, THE AUXILIARY BRANCH DENOTES THE NON-SSM BRANCH, AND THE FUSED FEATURE REPRESENTS THE FINAL OUTPUT OF DPSF OBTAINED BY INTEGRATING THE TWO BRANCHES

Feature source	Datasets		
	SV	IP	PU
Auxiliary branch	7.93	13.78	9.82
SSM branch	7.89	9.92	6.22
Fused feature	14.2	22.9	7.45

As shown in Table VII, the FR values of the auxiliary branch are higher than those of the SSM branch on all three datasets. Particularly, on the IP and PU datasets, the auxiliary branch achieves FR improvements of 3.86 and 3.60 over the SSM branch, respectively. This indicates that the auxiliary branch provides more discriminative local representations, while some fine-grained local variations may be weakened during long-range dependency modeling in the SSM branch. Specifically, the SSM branch focuses on global contextual dependency modeling, while the auxiliary branch enhances local spatial–spectral responses through convolution operations, providing more discriminative feature representations.

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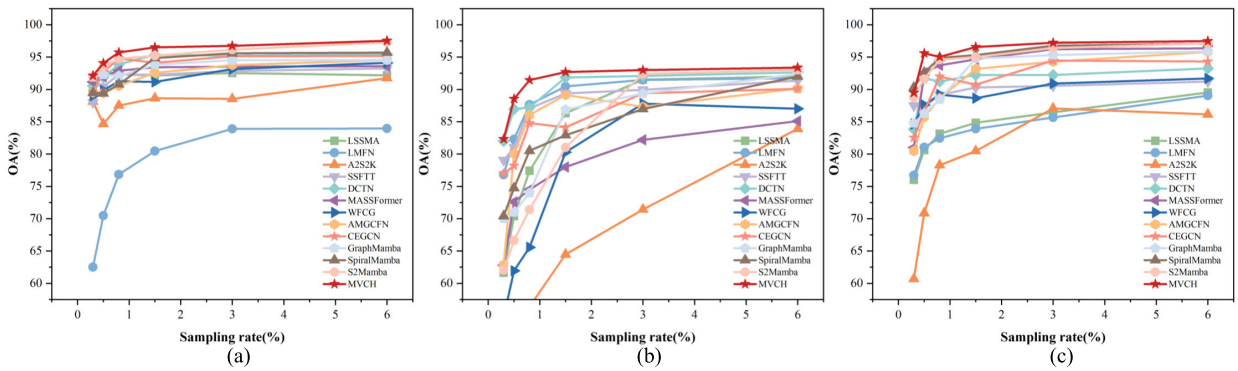


Fig. 10. OA of different methods under different sampling ratios on three datasets. (a) SV dataset. (b) IP dataset. (c) PU dataset.

TABLE VIII

MODEL COMPLEXITY OF THE PROPOSED MVCH AND COMPARISON METHODS. (BOLD INDICATES OUR RESULTS)

Dataset	Metric	Methods												
		LSSMA	LMFN	A2S2K	SSFTTNet	DCTN	MASSFormer	WFCG	AMGCFN	CEGCN	GraphMamba	SpiralMamba	S2Mamba	MVCH
SV	Train(S)	75.16	58.16	32.66	40.47	36.61	6.40	48.39	9.22	17.64	2.40	5.12	3.40	**36.50**
	Test(S)	10.65	5.92	7.63	5.00	28.63	1.34	1.68	1.72	0.10	0.29	4.96	1.44	**4.41**
	Flops(G)	5.48	0.47	6.13	2.34	75.00	0.80	8.62	10.92	8.58	3.66	11.90	1.30	**4.48**
	Param(M)	0.01	0.04	0.23	0.16	45.28	0.31	0.08	0.34	0.17	0.31	2.14	0.13	**0.29**
IP	Train(S)	13.30	4.35	6.04	9.60	15.78	4.49	6.45	3.89	3.02	5.50	1.70	0.74	**5.14**
	Test(S)	1.79	0.82	1.29	0.66	4.80	0.26	0.32	0.10	0.41	0.42	0.83	0.59	**0.76**
	Flops(G)	3.01	0.20	6.06	0.10	75.00	1.25	1.62	2.05	1.61	3.09	11.53	1.19	**4.41**
	Param(M)	0.01	0.04	0.22	0.15	45.30	0.31	0.08	0.34	0.17	0.82	2.06	0.13	**0.28**
PU	Train(S)	35.91	26.31	22.08	28.72	29.14	5.93	592.35	33.98	16.36	11.01	2.68	2.00	**19.35**
	Test(S)	9.86	6.71	10.62	6.24	35.28	1.10	2.89	0.20	3.21	1.69	2.34	1.67	**5.12**
	Flops(G)	2.73	0.26	4.46	2.26	49.56	0.33	13.22	15.01	13.07	3.65	3.07	1.19	**2.67**
	Param(M)	0.01	0.02	0.17	0.15	39.09	0.31	0.06	0.32	0.15	0.31	0.56	0.12	**0.17**

Furthermore, the fused features achieve higher FR values than the SSM branch on all datasets, demonstrating that the local discriminative information from the auxiliary branch can complement the global contextual information through dual-patch fusion. It is worth noting that the fused feature obtains a lower FR value than the auxiliary branch on the PU dataset. This is because the fused representation integrates both global contextual information and local discriminative cues, balancing global semantic consistency and local feature discrimination. Nevertheless, the fused features still achieve higher FR values than the SSM branch, confirming the complementary contribution of the auxiliary branch to feature discrimination.

G. Performance Under Limited Samples

Existing studies on HSI classification [52], [53] commonly employ low sampling rates to construct limited-label settings and analyze model performance with varying numbers of training samples. Following this experimental protocol, we report the OA of different methods under different sampling rates on the three datasets. As shown in Fig. 10, 0.3%, 0.5%, 0.8%, 1.5%, 3%, and 6% of the samples from each class are randomly selected for training, while the remaining samples are used for testing. It can be observed that the overall classification accuracy of all methods generally increases with the sampling rate, while the performance differences among methods are more pronounced at lower sampling rates, particularly under the limited-label settings of 0.3%–0.8%. As the sampling rate further increases, the performance gaps among methods gradually narrow. Based on these comparisons across multiple sampling rates, and considering our objective of evaluating different methods under limited-label conditions using a low and unified training ratio, 0.8% is empirically selected as the default sampling rate for subsequent experiments.

H. Model Complexity Analysis

Table VIII shows that different methods exhibit notable differences in parameter scale, computational efficiency, and classification performance. Some CNN-based approaches (e.g., LSSMA and LMFN), although involving relatively fewer parameters, rely on multilayer convolution operations, which introduce high computational complexity when processing high-dimensional hyperspectral data, resulting in significantly increased training and testing time. Transformer-based methods (e.g., DCTN and MASSFormer) demonstrate advantages in global feature modeling and long-range dependency capture. However, these benefits are often accompanied by increased model complexity and computational overhead. Taking DCTN as an example, its FLOPs on the SV dataset reach 75 G, with a parameter count of 45.28 M and a testing time of 28.63 s. In contrast, MVCH requires only 4.48 G FLOPs, 0.29 M parameters, and 4.41 s testing time on the SV dataset, while still maintaining competitive classification accuracy Table III,

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indicating a substantial reduction in computational cost and model size. In addition, SSM-based methods generally exhibit faster training and inference speeds. S2Mamba is characterized by the smallest parameter size and lower computational cost, while SpiralMamba maintains relatively high efficiency with a larger parameter scale. Although GCN-based methods generally exhibit moderate parameter scales and time consumption, their FLOPs on the SV and PU datasets are relatively high. This suggests that graph construction and iterative graph convolution operations introduce additional computational burdens, which limit further improvements in resource efficiency.

Compared with these methods, MVCH maintains high classification accuracy while keeping training time, testing time, parameter size, and FLOPs within reasonable ranges, demonstrating an effective balance between hyperspectral feature representation and computational efficiency with competitive performance at relatively low cost.

V. Conclusion

In this work, a novel MVCH framework is proposed for HSI classification. MVCH establishes a unified multiview representation framework by collaboratively modeling complementary dependency views, including structural relational, local spatial–spectral, global contextual, local perceptual, and dynamic dependency views. Specifically, GDC jointly exploits structural relational and local spatial–spectral views to enhance the representation of fine-grained feature variations and discontinuous structural regions. Meanwhile, CIT integrates global contextual, local perceptual, and dynamic dependency views to capture complementary contextual and dynamic dependency information. Through the interaction of these complementary views within a unified representation space, MVCH achieves a more comprehensive dependency representation for HSI classification. Extensive experiments on three representative HSI datasets demonstrate that MVCH consistently achieves favorable performance in terms of classification accuracy and efficiency. Notably, the proposed framework outperforms several representative methods under limited training sample scenarios, demonstrating strong generalization capability.

In future work, further investigations will focus on extending MVCH to larger-scale hyperspectral scenes and cross-sensor datasets. Moreover, more efficient structured relationship modeling strategies will be explored to reduce computational overhead. In addition, the integration of self-supervised learning or multimodal information fusion is expected to further enhance representation capability and practical applicability under extremely limited training data scenarios.

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